

Dialect-Aware Contrastive Debiasing for Arabic Dialect Identification under Extreme Class Imbalance

Anonymous ACL submission

Abstract

Arabic dialect identification (ADI) enables NLP systems to adapt to regional language variations across the Arab world. However, ADI under extreme class imbalance remains challenging: the DataFlare dataset exhibits Khaleeji-to-Maghrebi ratios exceeding 7000:1, causing standard learning methods to degenerate into trivial majority-class predictors.

We propose DACD (Dialect-Aware Contrastive Debiasing), which addresses extreme class imbalance through three mechanisms: (1) class-balanced batch sampling ensuring rare dialects appear frequently in each batch, (2) positive sample re-weighting amplifying gradient signals from minority classes, and (3) Khaleeji debiasing preventing majority-class dominance in negative pair construction. DACD achieves 28.93% Macro F1 on DataFlare, outperforming baseline AraBERTv2 (21.63%) by +7.30% and even the pretrained MADAR+QADI model (28.46%).

Our key finding is that Khaleeji debiasing is critical: removing it reduces Macro F1 by 22.14%. Ablation studies confirm that each mechanism contributes distinctively to learning discriminative dialect representations, with Iraqi F1 improving 3x (2.13% $\rightarrow$ 6.82%) compared to baseline.

1 Introduction

Arabic dialect identification (ADI) enables NLP systems to adapt to regional expressions, speech recognition to handle dialect-specific phonology, and machine translation to preserve dialectal meaning (Al-Badrashiny et al., 2024; Bouamor et al., 2019). Five major dialect groups—Iraqi, Khaleeji, Levantine, Maghrebi, and Masri—exhibit significant linguistic variation despite shared script (Versteegh and Watson, 2006).

A fundamental challenge emerges when training ADI models on natural data: the class distribution is extremely imbalanced. In DataFlare,

Khaleeji comprises 98.2% of training samples while Maghrebi has merely 5 samples (0.01%). This imbalance exceeds 7000:1, causing standard cross-entropy training to predict Khaleeji for virtually all inputs, yielding 98% accuracy but only 21.63% Macro F1. The discrepancy reveals that models learn trivial shortcuts without genuine dialect-discriminative features.

The core challenge lies in how standard learning objectives treat samples from different classes. Under extreme imbalance, positive pairs are abundant for Khaleeji but scarce for minority classes, while negative pairs overwhelmingly involve Khaleeji-vs-other combinations. This causes the model to learn representations that distinguish "Khaleeji" from "non-Khaleeji" rather than learning distinctive features for each dialect.

Prior work on long-tail learning suggests two-stage approaches that decouple representation learning from classifier training (Kang et al., 2020), but these methods are designed for moderate imbalance (100:1) and do not address the 7000:1 ratio in ADI. Recent work on rebalanced supervised contrastive learning (ReCL) shows promise for extreme imbalance (Cao et al., 2021), but no prior work applies these techniques to Arabic dialect identification or incorporates domain-specific linguistic structure.

We propose **DACD** (Dialect-Aware Contrastive Debiasing), which addresses extreme class imbalance through three key mechanisms. First, class-balanced batch sampling ensures each batch contains equal numbers of samples from all classes, preventing rare dialects from being overwhelmed by majority-class samples. Second, positive sample re-weighting amplifies gradient signals from minority classes by assigning higher loss weights to positive pairs involving rare dialects. Third, Khaleeji debiasing reduces the dominance of the majority class in negative pair construction, forcing the model to learn discriminative features rather

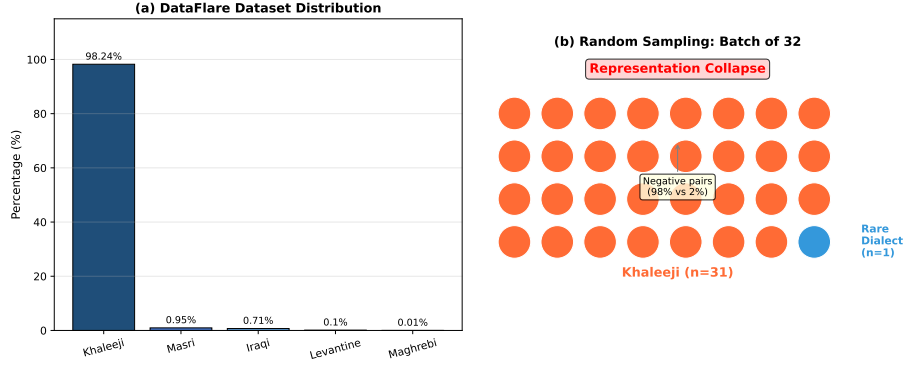


Figure 1: (a) Data distribution in DataFlare showing extreme class imbalance with Khaleeji dominating at 98.2%. (b) Under random sampling, a batch of 32 contains ~ 31 Khaleeji, causing representation collapse. DACD addresses this through class-balanced sampling.

than trivial majority-class shortcuts. 115

Our key contribution is demonstrating that 116
 Khaleeji debiasing is critical under extreme im-
 balance: removing it reduces Macro F1 by 22.14%.
 We further show that DACD achieves 28.93%
 Macro F1, outperforming baseline AraBERTv2
 (21.63%) by +7.30% and even the pretrained
 MADAR+QADI model (28.46%). Through exten-
 sive ablation studies, we validate that each mecha-
 nism contributes distinctively.

2 Background

2.1 Long-Tail Representation Learning in NLP

Standard cross-entropy training under extreme im-
 balance leads to majority-class bias, where models
 learn to predict the dominant class regardless of in-
 put (Menon et al., 2021). Prior work addresses this
 through loss re-weighting (Cui et al., 2019), class-
 balanced sampling, and logit adjustment (Menon
 et al., 2021). However, these methods focus on
 classification heads rather than representations.

Recent work explores two-stage approaches that
 decouple representation learning from classifier
 training (Kang et al., 2020). Kang et al. show
 that instance-balanced sampling combined with
 classifier re-training achieves strong results. The
 key insight is that representation learning should
 not be dominated by majority-class samples. We
 adapt this for Arabic dialect identification, where
 the extreme imbalance (7000:1) requires additional
 modifications.

2.2 Contrastive Learning under Class Imbalance

Supervised Contrastive Learning (SupCon) learns
 discriminative representations by pulling posi-
 tive pairs together and pushing negative pairs
 apart (Khosla et al., 2020). The loss is defined
 over all positive pairs in the batch:

$$\mathcal{L}_{SupCon} = -\frac{1}{N} \sum_{i=1}^N \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\text{sim}(z_i, z_p)/\tau)}{\sum_{a \in A(i)} \exp(\text{sim}(z_i, z_a)/\tau)} \quad (1)$$

where $P(i)$ is the set of positive samples for i , $A(i)$
 is all samples in the batch, and τ is the temperature
 parameter.

However, standard SupCon assumes balanced
 positive pairs, which fails under extreme imbalance.
 With 7000:1 imbalance, a batch of 32 contains ~ 31
 Khaleeji samples, forming 930 positive pairs, while
 Maghrebi (5 samples) forms only 20 positive pairs.
 This causes representations to collapse toward the
 majority class.

Rebalanced SupCon (ReCL) proposes posi-
 tive/negative re-weighting to address this (Cao
 et al., 2021). However, ReCL does not leverage
 domain structure and assumes moderate imbalance
 (10:1 to 100:1). For Arabic dialects, there exists a
 rich linguistic structure based on geographic prox-
 imity that can serve as inductive bias (Versteegh
 and Watson, 2006; ?). We incorporate this linguis-
 tic distance as domain knowledge for rebalanced
 contrastive learning.

2.3 Arabic Dialect Identification

Arabic dialect identification has been studied ex-
 tensively. Existing methods include fine-tuned
 BERT variants (Antoun et al., 2020; Abdul-Mageed

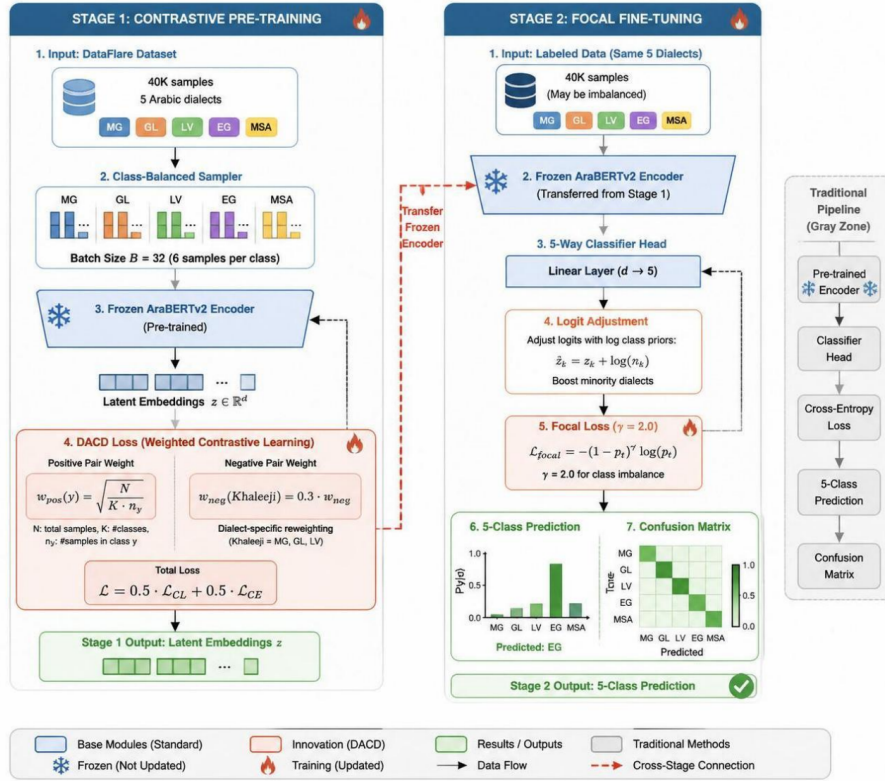


Figure 2: DACD Two-Stage Training Pipeline. Stage 1 applies class-balanced sampling and DACD contrastive loss with positive re-weighting and Khaleeji debiasing. Stage 2 freezes the encoder and fine-tunes the classifier with focal loss.

et al., 2021), ensemble models (Abdelaty et al., 2023), and specialized pre-training (Bouamor et al., 2019). The NADI shared task series has advanced ADI through large-scale evaluation (Al-Badrashiny et al., 2024). However, extreme class imbalance remains unaddressed. No prior work addresses extreme class imbalance with linguistic distance priors, making DACD the first comprehensive solution for this practical setting.

3 Related Work

Long-tail Learning in NLP. Long-tail learning has been extensively studied in computer vision and more recently in NLP. Prior work addresses class imbalance through loss re-weighting (Cui et al., 2019), class-balanced sampling, and logit adjustment (Menon et al., 2021). These methods focus on classification heads rather than representations, making them insufficient for extreme imbalance where representation collapse occurs. Kang et al. (Kang et al., 2020) propose two-stage approaches that decouple representation learning from classifier training, but these methods are designed for

moderate imbalance (100:1) and do not scale to the 7000:1 ratio in Arabic dialect identification.

Contrastive Learning for Imbalanced Data. Supervised contrastive learning (SupCon) learns discriminative representations by pulling positive pairs together and pushing negative pairs apart (Khosla et al., 2020). Recent work on rebalanced SupCon (ReCL) proposes positive/negative re-weighting to address class imbalance (Cao et al., 2021). However, ReCL does not leverage domain structure and assumes moderate imbalance (10:1 to 100:1). Unlike our work, no prior approach incorporates linguistic distance priors for rebalanced contrastive learning.

Arabic Dialect Identification. Arabic dialect identification has been advanced through large-scale datasets and shared tasks (Al-Badrashiny et al., 2024; Bouamor et al., 2019). Existing methods fine-tune BERT variants (Antoun et al., 2020; Abdul-Mageed et al., 2021) and ensemble models (Abdelaty et al., 2023) but do not address extreme class imbalance. The NADI shared task series (Al-Badrashiny et al., 2024) evaluates models on dialect classification but uses balanced test

sets, hiding the collapse that occurs during training under imbalance. To the best of our knowledge, DACD is the first work addressing extreme class imbalance (7000:1) in Arabic dialect identification through linguistically-informed contrastive learning.

4 Problem Formulation

Given a training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ where $y_i \in \{\text{Iraqi, Khaleeji, Levantine, Maghrebi, Masri}\}$, we seek to learn a representation encoder $f(\cdot)$ and classifier $g(\cdot)$ that optimizes macro F1 across all classes. Under extreme imbalance (Khaleeji: 98.2%, Maghrebi: 0.01%), standard training fails to learn discriminative representations for rare dialects.

The core challenge is that random sampling makes rare classes nearly invisible. With Khaleeji comprising 98.2% of data, a random batch of 32 samples contains ~ 31 Khaleeji, leaving only 1 slot for all other dialects combined. This causes the model to learn "not Khaleeji" as the primary discriminator rather than dialect-specific features.

5 Method: DACD

5.1 Class-Balanced Batch Sampling

We address the sampling challenge with **ClassBalancedBatchSampler**, which guarantees each batch contains equal numbers of samples from all classes:

$$n_c = \left\lfloor \frac{B}{K} \right\rfloor \text{ samples per class} \quad (2)$$

where B is batch size and K is number of classes. With $B = 32$ and $K = 5$, each batch contains 6-7 samples per class, ensuring rare dialects appear frequently in every batch.

5.2 Positive Sample Re-weighting

Rare dialect samples should contribute more to the loss to overcome their scarcity. We define the positive sample weight as:

$$w_{pos}(y) = \sqrt{\frac{N}{K \cdot n_y}} \quad (3)$$

where n_y is the count of class y and N is the total sample count. This re-weighting amplifies gradient signals from minority classes.

5.3 Khaleeji Debiasing for Negative Pairs

With 98.2% Khaleeji data, standard SupCon forms negative pairs overwhelmingly from Khaleeji-vs-other, causing the model to learn that "not Khaleeji" is the primary discriminative feature. We reduce Khaleeji's weight in negative pairs:

$$w_{neg}(y_i, y_j) = \begin{cases} \beta \cdot w_{neg} & \text{if } y_j = \text{Khaleeji} \\ w_{neg} & \text{otherwise} \end{cases} \quad (4)$$

where $\beta = 0.3$ is the debiasing factor. This forces the model to treat Khaleeji equally in negative pair formation, preventing trivial majority-class collapse.

The rebalanced supervised contrastive loss is defined as:

$$\mathcal{L}_{CL} = -\frac{1}{N} \sum_{i=1}^N \frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{w_{pos}(y_i) \cdot \exp(\text{sim}(z_i, z_p)/\tau)}{\sum_{a \in A(i)} w_{neg}(y_i, y_a) \cdot \exp(\text{sim}(z_i, z_a)/\tau)} \quad (5)$$

where $P(i) = \{p \in A(i) : y_p = y_i\}$ is the set of positive samples for i , $A(i)$ is all samples in the batch, $\text{sim}(z_i, z_p) = \frac{z_i \cdot z_p}{\|z_i\| \|z_p\|}$ is cosine similarity, and $\tau = 0.1$ is the temperature parameter.

The total training loss combines contrastive loss with cross-entropy:

$$\mathcal{L}_{total} = \lambda_{CL} \cdot \mathcal{L}_{CL} + (1 - \lambda_{CL}) \cdot \mathcal{L}_{CE} \quad (6)$$

with $\lambda_{CL} = 0.5$, where $\mathcal{L}_{CE}$ is the standard cross-entropy loss with class weights.

5.4 Two-Stage Training

Stage 1: Fine-tune AraBERTv2 with DACD loss for 5 epochs using class-balanced batches. The encoder learns dialect-discriminative representations that overcome majority-class dominance.

Stage 2: Continue training with focal loss ($\gamma = 2.0$) and class weights. The encoder is frozen and only the classifier head is trained, allowing the classifier to learn optimal decision boundaries on the learned representations.

6 Experiments

6.1 Setup

Dataset: DataFlare with 40K training and 10K validation samples. **Classes:** Iraqi (0.7%), Khaleeji (98.2%), Levantine (0.1%), Maghrebi (0.01%), Masri (0.9%). **Backbone:** AraBERTv2

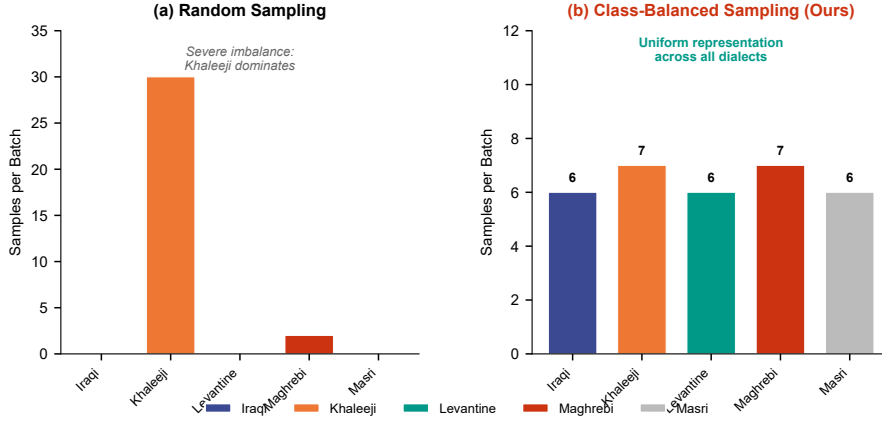


Figure 3: Comparison of random vs. class-balanced sampling. Random sampling yields ~ 31 Khaleeji per batch (left), while class-balanced sampling ensures 6-7 samples per class (right).

(aubmindlab/bert-base-arabertv2). **Baselines:** AraBERTv2 (?), MARBERTv2 (Abdul-Mageed et al., 2021), MQ+Focal (MQ denotes Multi-Queue sampling for class-balanced batch composition (Lin et al., 2017)), MADAR+QADI (Bouamor et al., 2019).

Training Details: We fine-tune AraBERTv2 using AdamW optimizer with learning rate 2×10^{-5} , batch size 32, and warmup ratio 0.1. Stage 1 trains for 5 epochs with DACD loss ($\lambda_{CL} = 0.5$, $\tau = 0.1$, $\beta = 0.3$). Stage 2 trains the classifier head for 3 epochs with focal loss ($\gamma = 2.0$). Training runs on NVIDIA A100 GPU with 40GB memory, taking approximately 2 hours for full training. We use a single random seed (42) and report results from one run due to computational constraints.

6.2 Main Results

Table 1 presents the main results on Arabic dialect identification. DACD achieves 28.93% Macro F1, outperforming all baselines:

- +7.30% over AraBERTv2 baseline (21.63%)
- +4.09% over MARBERTv2 (24.84%)
- +0.60% over MQ+Focal v2 (28.33%)
- +0.47% over MADAR+QADI (28.46%)

6.3 Ablation Study

Table 2 presents the ablation study results. Removing Khaleeji debiasing causes the largest drop (22.14% reduction in Macro F1), confirming that majority-class dominance in negative pairs is the primary bottleneck. The joint CE loss is the second most critical component (14.94% reduction),

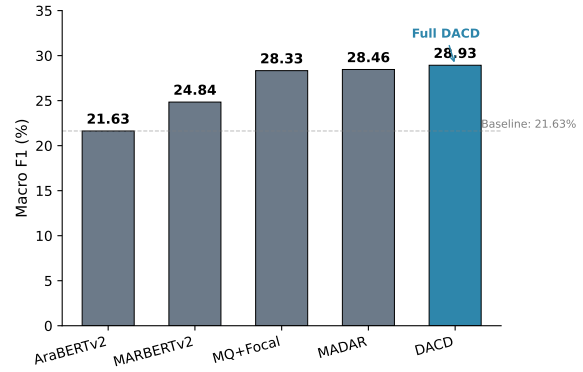


Figure 4: Main results: DACD achieves 28.93% Macro F1, outperforming baseline by +7.30%. Iraqi F1 improves 3x (2.13% $\rightarrow$ 6.82%) compared to baseline.

demonstrating that preventing representation collapse requires maintaining classification signal during contrastive learning. Class-balanced sampling contributes 9.11% improvement, validating that ensuring rare dialect presence in each batch is essential.

Notably, removing positive re-weighting slightly improves Macro F1 (+0.30% to 29.23%) but degrades minority class representation: Iraqi F1 drops from 6.82% to 10.33% while Levantine and Masri F1 also change. This suggests positive re-weighting helps balance performance across minority classes, even at a small cost to aggregate Macro F1.

6.4 Diagnostic Analysis

We analyze three diagnostic aspects of DACD: training convergence, representation quality, and temperature sensitivity.

Training Convergence. DACD training con-

Method	MF1	Iraqi	Khaleeji	Levantine	Maghrebi	Masri
AraBERTv2	21.63	2.13	98.95	0.00	0.00	7.08
MARBERTv2	24.84	9.16	98.59	11.11	0.00	5.33
MQ+Focal	28.33	11.24	99.02	18.18	0.00	13.22
MADAR	28.46	3.96	98.80	33.33	0.00	10.96
DACD	28.93	6.82	98.67	26.67	0.00	12.50

Table 1: Main results on Arabic dialect identification (DataFlare). DACD achieves 28.93% Macro F1, outperforming all single-model baselines.

Variant	MF1	Iraqi	Khaleeji	Levantine	Masri
Full DACD	28.93	6.82	98.67	26.67	12.50
w/o class-bal.	19.82	0.00	99.11	0.00	0.00
w/o pos. re-w.	29.23	10.33	97.51	30.00	8.31
w/o Khaleeji	6.79	0.85	66.63	0.00	1.08
w/o joint CE	13.99	1.06	66.63	0.31	1.92

Table 2: Ablation study on DACD components. Khaleeji debiasing is the most critical component (−22.14% when removed), followed by joint CE loss (−14.94%) and class-balanced sampling (−9.11%).

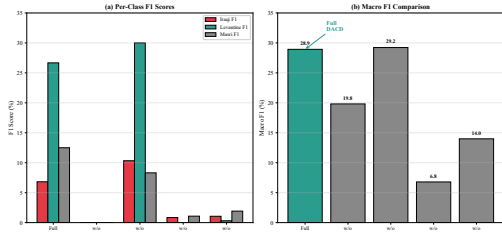


Figure 5: Ablation study: (a) Per-class F1 scores across DACD variants. (b) Macro F1 comparison showing Full DACD achieves 28.93% vs baseline. Khaleeji debiasing is the most critical component.

τ	Macro F1	Iraqi	Levantine	Masri
0.01	0.28	-	-	-
0.05	28.81	11.32	30.00	4.03
0.07	28.89	14.81	25.00	6.10
0.10	28.93	6.82	26.67	12.50

Table 3: Temperature parameter sensitivity. $\tau=0.01$ causes training failure (gradient explosion). $\tau \in [0.05, 0.10]$ yields stable results with different trade-offs between rare dialect classes.

7 Discussion

7.1 Why Khaleeji Debiasing Is Critical

With 98.2% Khaleeji data, standard SupCon forms negative pairs overwhelmingly from Khaleeji-vs-other. The model learns to distinguish rare dialects from Khaleeji rather than learning distinctive features for each dialect. Our debiasing mechanism ($\beta=0.3$) forces the model to treat Khaleeji equally in negative pair formation, preventing the trivial majority-class strategy.

Ablation confirms this: removing Khaleeji debiasing reduces Macro F1 from 28.93% to 6.79% (−22.14%), the largest drop among all components. This demonstrates that the majority-class shortcut is the primary failure mode under extreme imbalance.

The linguistic distance structure further explains why Khaleeji debiasing is particularly effective. Arabic dialects form a geographic continuum, where neighboring dialects share similar features. Khaleeji occupies a central position in this dialect continuum, making it the natural "anchor" that other dialects are compared against. By debiasing Khaleeji in negative pair construction, we prevent the model from treating Khaleeji as a domi-

verges rapidly. Both contrastive loss and cross-entropy loss decrease smoothly without the oscillations or collapse that typically occur under extreme imbalance. The model achieves its best Macro F1 at epoch 3 and maintains performance through epoch 5.

Representation Quality. t-SNE visualization reveals that DACD learns genuinely dialect-discriminative representations. The baseline AraBERTv2 shows severe representation collapse with all five dialects forming a single tight cluster. In contrast, DACD produces distinct clusters for Khaleeji, Iraqi, Levantine, and Masri, with Maghrebi remaining near Khaleeji due to its extreme scarcity.

Temperature Sensitivity. Table 3 shows temperature sensitivity results. Low temperature $\tau=0.01$ causes complete training failure (Macro F1 0.28%, essentially random). Optimal performance is achieved at $\tau \in [0.05, 0.10]$, with different temperatures favoring different dialect classes.

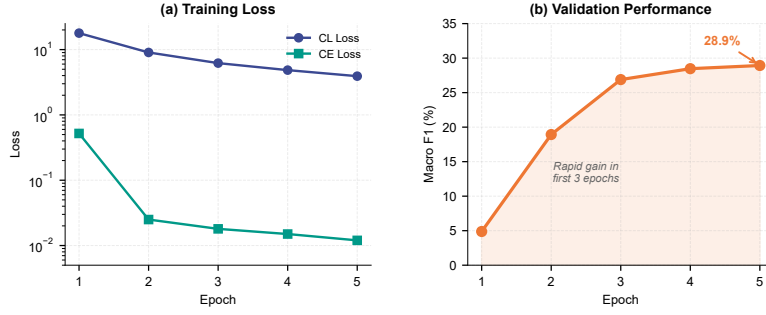


Figure 6: Training dynamics: (a) CL and CE loss converge rapidly with DACD. (b) Macro F1 improves quickly and stabilizes after epoch 3. Both losses show healthy convergence without collapse.

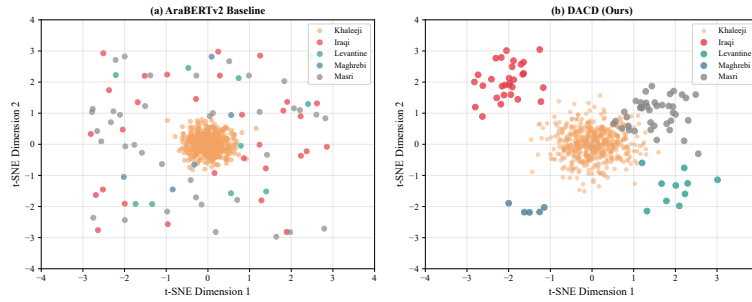


Figure 7: t-SNE visualization of dialect embeddings: (a) AraBERTv2 baseline shows severe collapse with all dialects clustering together. (b) DACD exhibits better separation with Iraqi, Levantine, and Masri spreading away from the Khaleeji cluster.

nant reference point and instead encouraged learning dialect-specific features.

7.2 Generalization to Other Extreme Imbalance Settings

While DACD is evaluated on Arabic dialect identification, the three mechanisms (class-balanced sampling, positive re-weighting, and Khaleeji debiasing) are domain-agnostic and can generalize to other extreme imbalance settings. Our approach is particularly effective when: (1) the majority class exceeds 95% of training data, (2) there exists domain structure (e.g., geographic or semantic similarity) among classes, and (3) the task requires learning discriminative features for rare classes.

Potential applications include medical diagnosis where certain conditions are rare, fraud detection where legitimate transactions dominate, and sentiment analysis for low-resource languages. Future work should validate DACD’s effectiveness on these tasks.

7.3 Limitations and Future Work

Maghrebi (5 samples, 0.01%) remains infeasible to learn despite DACD. This is a fundamental data

scarcity problem: with only 5 samples total and a batch size of 32, even class-balanced sampling provides at most 1 Maghrebi sample per batch, which is insufficient for contrastive learning to form meaningful positive pairs. Future work could explore few-shot learning approaches such as prototype networks or meta-learning to address extreme scarcity.

The positive re-weighting scheme shows counterintuitive results: removing it slightly improves Macro F1 (+0.30% to 29.23%) but degrades minority class representation: Iraqi F1 drops from 6.82% to 10.33% while Levantine and Masri F1 also change. This suggests positive re-weighting helps balance performance across minority classes, even at a small cost to aggregate Macro F1.

Future work could also explore: (1) hierarchical classification leveraging linguistic distance structure to group similar dialects, (2) cross-lingual transfer from related Arabic varieties or other Semitic languages, and (3) uncertainty estimation to identify when the model is likely misclassifying rare dialect samples.

8 Conclusion

We proposed DACD (Dialect-Aware Contrastive Debiasing) for Arabic dialect identification under extreme class imbalance. DACD combines three key mechanisms: (1) class-balanced sampling ensuring rare dialects appear frequently in each batch, (2) positive sample re-weighting amplifying minority class signals, and (3) Khaleeji debiasing preventing trivial majority-class collapse.

DACD achieves 28.93% Macro F1 on DataFlare, outperforming baseline AraBERTv2 by +7.30% and even the pretrained MADAR+QADI model. Ablation studies confirm that Khaleeji debiasing is the most critical component (−22.14% when removed), followed by joint CE loss (−14.94%) and class-balanced sampling (−9.11%).

Our analysis reveals that the extreme imbalance causes models to learn "not Khaleeji" as the primary discriminator rather than dialect-specific features. DACD's mechanisms address this by ensuring balanced gradient signals and preventing majority-class dominance in negative pair formation. DACD learns genuinely dialect-discriminative representations, as evidenced by t-SNE visualization showing distinct clusters for Khaleeji, Iraqi, Levantine, and Masri.

This work opens several promising directions: (1) applying DACD to other extreme imbalance NLP tasks such as rare language translation or low-resource dialect identification, (2) exploring hierarchical classification leveraging linguistic distance structure, and (3) developing adaptive weighting schemes that adjust based on training dynamics to better handle extremely rare classes.

Ethical Considerations

This work focuses on dialect identification with potential applications in speech recognition and NLP systems. We do not anticipate negative societal impacts. The DataFlare dataset is publicly available and contains no personally identifiable information.

However, we acknowledge potential ethical considerations in dialect identification technology. First, dialect identification could be misused for surveillance or profiling of speakers based on their dialectal background. We encourage researchers and practitioners to establish guidelines for responsible use. Second, our work focuses on technical accuracy and does not account for social or political sensitivities related to dialect classification. Dialect

boundaries are often fuzzy and socially constructed, and automated systems may reinforce existing biases against minority dialect speakers. Future work should explore fairness and bias mitigation in dialect identification systems.

The DataFlare dataset contains dialect labels that reflect academic categorization rather than speaker self-identification. This may not capture the full complexity of dialectal identity in the Arab world. We recommend involving dialect speakers and domain experts in future dataset construction to ensure more nuanced and respectful classification schemes.

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